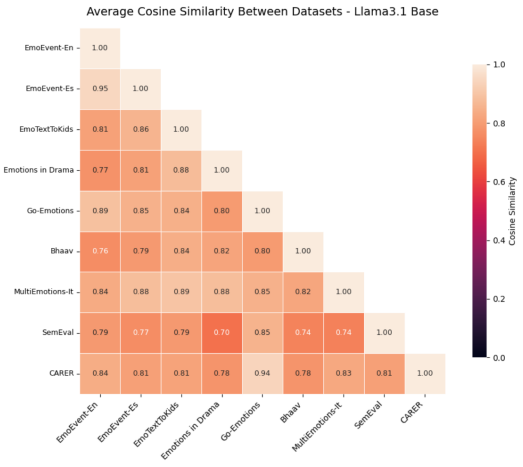

Figure 4: Cosine similarity of emotional centroids between datasets for Llama3.1-Base.

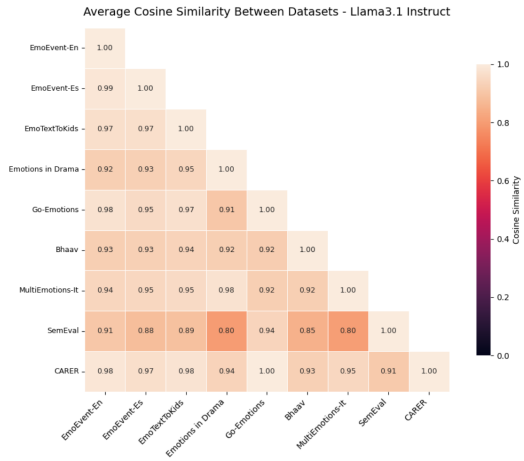

Figure 5: Cosine similarity of emotional centroids between datasets for Llama3.1-Instruct.

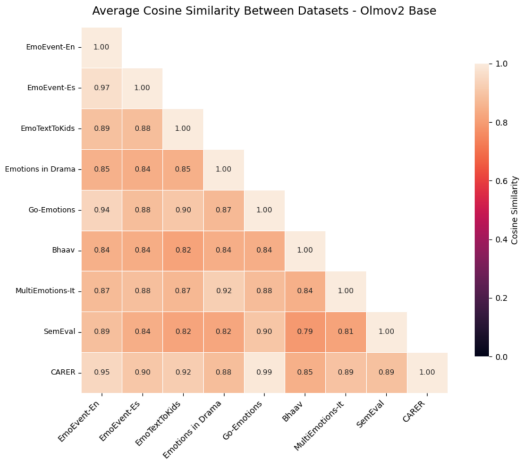

Figure 6: Cosine similarity of emotional centroids between datasets for Olmov2-Base.

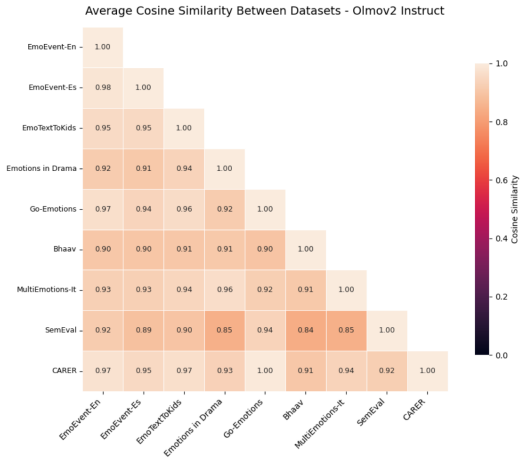

Figure 7: Cosine similarity of emotional centroids between datasets for Olmov2-Instruct.

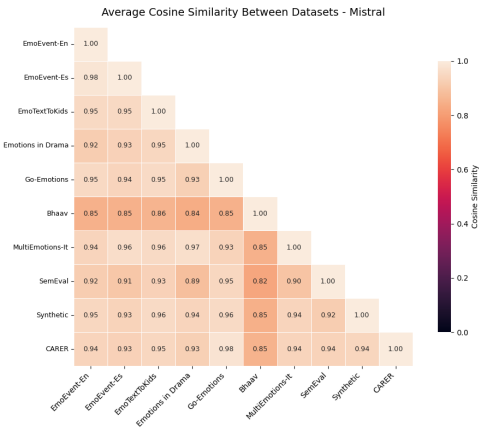

Figure 8: Cosine similarity of emotional centroids between datasets for Ministral.

D Distortion Score Contextualization

To situate the stress and distortion scores reported in section 4, we tested the alignment between Llama3.1-8B-Instruct's synthetic emotion subspace and its activations on GSM8K. Although GSM8K is affect-neutral and task-oriented, "neutral" language is itself an emotional register; a truly emotionless baseline is impossible. Still, GSM8K provides a controlled case expected to align poorly with emotional structure. Consistent with this, it yields the largest distortions and among the highest stress values: Stress-1 = 0.517 ± 0.130, Stress-2 = 0.284 ± 0.244, Sammon stress = 22.84 ± 32.41, average distortion = 29.06 ± 36.42, L2 distortion = $134{,}869 \pm 168{,}270$, and sigma distortion = $14{,}435{,}586 \pm 11{,}306{,}312$.

These results confirm that alignment metrics only hold for datasets with consistent emotional structure: GSM8K, though emotionally neutral, is misaligned with the emotional manifold. Together with the lexical and syntactic diversity analysis presented in Table 4, these findings provide strong evidence that the observed structure in our aligned datasets is not an artifact of surface-level features, but instead reflects a consistent and abstract encoding of emotion in the model's internal geometry.

E Universality Expanded

In Section 4, stress-2 and average distortion scores for the models are discussed, with results compared across English and non-English datasets. In this appendix, the corresponding per-dataset results for stress, distortion, and probe accuracy are presented, along with an expanded set of stress and distortion metrics.

Table 5 presents the stress and distortion score per-model and per-dataset. LLaMA-3.1-8B-Base, which achieves the lowest stress values across datasets. LLaMA-3.1-8B-Instruct and Ministral also yield relatively low Stress-1 and Stress-2 scores, though somewhat higher than the LLaMA-3.1-Base model, suggesting that instruction-tuned variants preserve emotional relations slightly less consistently. Interestingly, OLMo-v2 shows the opposite trend: the base model achieves lower stress than the instruction-tuned variant, but both versions exhibit substantially higher stress values than LLaMA or Ministral. This pattern suggests that OLMo encodes a less unified emotional space than the other models. Projection into the 50D synthetic subspace increases stress across all models, consistent with expected compression effects. For models with already low stress values, the increases remain modest and the geometry remains coherent, whereas for models with elevated stress (e.g., OLMo), dimensionality reduction does not mitigate distortion.

To complement the stress scores, three high-dimensional embedding distortion metrics are reported: average distortion, ℓ_2-distortion, and σ-distortion. These metrics highlight the contrast between base and instruction-tuned models. Both LLaMA-3.1-8B-Base and OLMo-v2-Base achieve distortion scores near ideal across most datasets (recall that no Ministral-Base model exists). For LLaMA-3.1-8B-Base, only two datasets deviate substantially from the ideal; excluding these outliers, its distortion metrics are more favorable than those of OLMo-v2-Base in both the full hidden-state space and the 50D synthetic subspace. See Appendix D for a contextualization of these scores.

The instruction-tuned models diverge considerably from their base counterparts. LLaMA-3.1-8B-Instruct produces roughly twice as many outlier datasets as LLaMA-3.1-8B-Base, while Ministral shows five outliers out of the nine datasets tested. Even after excluding these outliers, distortion remains higher for both models in both full-space and 50D-space analyses. Despite this, LLaMA-3.1-8B-Instruct and Ministral still maintain broadly consistent emotional structure across multiple datasets and languages. OLMov2-Instruct, however, performs markedly worse: the emotional spaces induced by different datasets are largely incompatible, with distortion elevated across nearly all layers and datasets.

Rather than reporting average distortion per layer for datasets with elevated distortion, we instead report the percentage of layers exhibiting high distortion. In LLaMA-3.1-8B-Base, only Go-Emotions shows substantial distortion (43% of layers). The instruction-tuned variant shows broader distortion on Go-Emotions, though across datasets a majority of layers are affected only about half the time. OLMo-v2-Instruct shows majority distortion in 3 of 9 datasets, with the rest under 50% (some under 40%). Ministral is most fragile: when distortion appears, it usually spans most layers, though fewer datasets are affected. Together, these results indicate that even when a model exhibits distortion in

Dataset	Stress-1 ↓	Stress-2 ↓	Sammon ↓	Avg Dist ↓	ℓ_2 ↓	σ ↓	Probe Acc. ↑
Llama3.1-8B-Base							
Go-Emotions	0.33 ± 0.33	0.13 ± 0.13			43%*		0.52 ± 0.52
CARER (Twitter)	0.38 ± 0.16	0.17 ± 0.15	0.18 ± 0.22	1.03 ± 0.24	1.11 ± 0.32	0.16 ± 0.33	0.35 ± 0.11
SemEval	0.34 ± 0.16	0.14 ± 0.15	0.14 ± 0.19	0.97 ± 0.23	1.03 ± 0.31	0.12 ± 0.26	0.46 ± 0.13
EmoEvent (EN)	0.36 ± 0.13	0.15 ± 0.13	0.15 ± 0.2	0.9 ± 0.19	0.97 ± 0.29	0.14 ± 0.26	0.55 ± 0.12
EmoEvent (ES)	0.38 ± 0.14	0.17 ± 0.14	0.16 ± 0.22	0.86 ± 0.2	0.93 ± 0.31	0.16 ± 0.29	0.5 ± 0.15
Bhaav (Hindi)	0.39 ± 0.15	0.17 ± 0.15	0.17 ± 0.19	0.86 ± 0.22	0.92 ± 0.31	0.16 ± 0.31	0.36 ± 0.1
German Drama	0.39 ± 0.17	0.18 ± 0.18	0.22 ± 0.47	1.06 ± 0.22	1.17 ± 0.48	0.48 ± 4.27	0.29 ± 0.1
MultiEmotions-It	0.38 ± 0.16	0.17 ± 0.15	0.19 ± 0.2	1.11 ± 0.24	1.19 ± 0.28	0.18 ± 0.48	0.46 ± 0.12
EmoTextToKids (FR)	0.41 ± 0.14	0.19 ± 0.15	0.19 ± 0.19	0.92 ± 0.24	0.99 ± 0.32	0.18 ± 0.33	0.39 ± 0.1
Average (Full-Space)	0.38 ± 0.15	0.17 ± 0.15	0.17 ± 0.25	0.97 ± 0.24	1.04 ± 0.35	0.2 ± 1.55	0.42 ± 0.14
Average (50D-Space)	0.58 ± 0.13	0.35 ± 0.16	0.33 ± 0.24	0.6 ± 0.21	0.68 ± 0.36	0.3 ± 1.66	0.39 ± 0.11
Llama3.1-8B-Instruct							
Go-Emotions	0.41 ± 0.11	0.18 ± 0.17			71%*		0.42 ± 0.06
CARER (Twitter)	0.4 ± 0.11	0.17 ± 0.11			40%*		0.58 ± 0.07
SemEval	0.6 ± 0.05	0.36 ± 0.06	0.32 ± 0.07	0.52 ± 0.09	0.56 ± 0.2	0.15 ± 0.28	0.49 ± 0.09
EmoEvent (EN)	0.4 ± 0.1	0.17 ± 0.09	0.2 ± 0.22	1.04 ± 0.14	1.15 ± 0.28	0.2 ± 0.3	0.11 ± 0.01
EmoEvent (ES)	0.43 ± 0.1	0.19 ± 0.1	0.22 ± 0.23	1.0 ± 0.14	1.12 ± 0.3	0.24 ± 0.38	0.11 ± 0.01
Bhaav (Hindi)	0.43 ± 0.11	0.2 ± 0.11	0.23 ± 0.3	1.02 ± 0.17	1.14 ± 0.38	0.23 ± 0.26	0.55 ± 0.05
German Drama	0.51 ± 0.15	0.29 ± 0.29			71%*		0.43 ± 0.07
MultiEmotions-It	0.49 ± 0.12	0.25 ± 0.13			46%*		0.57 ± 0.07
EmoTextToKids (FR)	0.42 ± 0.1	0.18 ± 0.1	0.21 ± 0.22	1.02 ± 0.13	1.13 ± 0.3	0.22 ± 0.27	0.60 ± 0.08
Average (Full-Space)	0.46 ± 0.12	0.22 ± 0.12	0.24 ± 0.23	0.92 ± 0.24	1.02 ± 0.37	0.21 ± 0.3	0.43 ± 0.19
Average (50D-Space)	0.55 ± 0.11	0.31 ± 0.13	0.33 ± 0.23	0.83 ± 0.27	0.96 ± 0.42	0.33 ± 0.35	0.37 ± 0.16
Olmov2-Base							
Go-Emotions	0.71 ± 0.36	0.63 ± 0.98	1.39 ± 5.32	1.65 ± 0.72	1.94 ± 1.08	0.34 ± 0.25	0.48 ± 0.06
CARER (Twitter)	0.66 ± 0.36	0.57 ± 1.03	1.22 ± 4.97	1.54 ± 0.67	1.8 ± 1.0	0.33 ± 0.25	0.61 ± 0.06
SemEval	0.78 ± 0.41	0.78 ± 0.95	1.35 ± 2.71	1.8 ± 0.63	2.02 ± 0.84	0.25 ± 0.18	0.6 ± 0.06
EmoEvent (EN)	0.56 ± 0.23	0.36 ± 0.42	0.75 ± 2.49	1.38 ± 0.5	1.61 ± 0.8	0.32 ± 0.24	0.11 ± 0.01
EmoEvent (ES)	0.55 ± 0.25	0.37 ± 0.5	0.76 ± 2.84	1.31 ± 0.59	1.53 ± 0.9	0.32 ± 0.22	0.11 ± 0.01
Bhaav (Hindi)	0.61 ± 0.26	0.44 ± 0.47	0.92 ± 2.86	1.26 ± 0.72	1.55 ± 1.09	0.44 ± 0.28	0.51 ± 0.05
German Drama	0.77 ± 0.58	0.93 ± 3.16	2.1 ± 12.92	1.42 ± 1.08	1.78 ± 1.53	0.52 ± 0.3	0.44 ± 0.05
MultiEmotions-It	0.74 ± 0.57	0.87 ± 2.74	1.88 ± 10.48	1.39 ± 1.04	1.71 ± 1.43	0.48 ± 0.28	0.58 ± 0.06
EmoTextToKids (FR)	0.6 ± 0.27	0.43 ± 0.59	0.87 ± 3.18	1.36 ± 0.47	1.62 ± 0.85	0.37 ± 0.33	0.59 ± 0.06
Average (Full-Space)	0.67 ± 0.4	0.6 ± 1.56	1.25 ± 6.43	1.46 ± 0.76	1.73 ± 1.1	0.37 ± 0.27	0.45 ± 0.2
Average (50D-Space)	0.72 ± 0.4	0.68 ± 1.61	1.56 ± 8.11	1.37 ± 0.9	1.78 ± 1.41	0.63 ± 0.42	0.34 ± 0.14
Olmov2-Instruct							
Go-Emotions	0.55 ± 0.19	0.34 ± 0.31			63%*		0.49 ± 0.06
CARER (Twitter)	0.57 ± 0.21	0.37 ± 0.41			38%*		0.65 ± 0.08
SemEval	0.46 ± 0.21	0.25 ± 0.3			47%*		0.61 ± 0.09
EmoEvent (EN)	0.51 ± 0.18	0.3 ± 0.26			41%*		0.11 ± 0.01
EmoEvent (ES)	0.51 ± 0.2	0.3 ± 0.3			41%*		0.11 ± 0.01
Bhaav (Hindi)	0.61 ± 0.27	0.44 ± 0.6			78%*		0.53 ± 0.05
German Drama	0.66 ± 0.29	0.52 ± 0.72			61%*		0.43 ± 0.05
MultiEmotions-It	0.65 ± 0.31	0.52 ± 0.88			41%*		0.56 ± 0.05
EmoTextToKids (FR)	0.56 ± 0.23	0.37 ± 0.45			34%*		0.62 ± 0.07
Average (Full-Space)	0.56 ± 0.24	0.38 ± 0.52			49%*		0.46 ± 0.20
Average (50D-Space)	0.65 ± 0.25	0.48 ± 0.58	979.03 ± 3086.26	882.19 ± 3055.12	482470.52 ± 978732.71	2673197.92 ± 3939472.37	0.36 ± 0.15
Ministral							
Go-Emotions	0.45 ± 0.17	0.23 ± 0.35			68%*		0.4 ± 0.05
CARER (Twitter)	0.47 ± 0.11	0.23 ± 0.11			70%*		0.49 ± 0.08
SemEval	0.38 ± 0.2	0.19 ± 0.51	0.24 ± 0.86	1.13 ± 0.19	1.22 ± 0.37	0.16 ± 0.49	0.55 ± 0.07
EmoEvent (EN)	0.41 ± 0.11	0.18 ± 0.1	0.21 ± 0.2	1.1 ± 0.14	1.2 ± 0.23	0.18 ± 0.15	0.11 ± 0.01
EmoEvent (ES)	0.48 ± 0.12	0.25 ± 0.12	0.32 ± 0.26	1.24 ± 0.14	1.37 ± 0.25	0.2 ± 0.16	0.11 ± 0.01
Bhaav (Hindi)	0.42 ± 0.11	0.19 ± 0.10			75%*		0.45 ± 0.05
German Drama	0.50 ± 0.25	0.32 ± 0.98			70%*		0.45 ± 0.07
MultiEmotions-It	0.49 ± 0.12	0.26 ± 0.14	0.31 ± 0.21	1.18 ± 0.11	1.31 ± 0.18	0.24 ± 0.26	0.66 ± 0.05
EmoTextToKids (FR)	0.43 ± 0.11	0.2 ± 0.11	0.24 ± 0.22	1.13 ± 0.12	1.24 ± 0.22	0.19 ± 0.16	0.57 ± 0.07
Average (Full-Space)	0.45 ± 0.16	0.23 ± 0.39	0.26 ± 0.44	1.16 ± 0.15	1.27 ± 0.27	0.19 ± 0.28	0.42 ± 0.19
Average (50D-Space)	0.52 ± 0.14	0.29 ± 0.4	0.32 ± 0.45	1.04 ± 0.18	1.19 ± 0.31	0.31 ± 0.3	0.36 ± 0.16

Table 5: Per-dataset distortion metrics and probe accuracy across three models. Lower distortion indicates greater geometric consistency. * in cells denotes very high stress/distortion. Instead of reporting the stress or distortion for that dataset, the percentage of layers that are highly distorted are reported.

parts of its architecture, large fractions of its layers still encode emotional spaces that remain universal across datasets, languages, and writing styles. These layers likely play a central role in maintaining consistent emotional representation within the model.

These per-dataset analyses show that while stress and distortion vary across model families and datasets, a broadly coherent emotional geometry is retained across datasets and languages, especially in the more stable base models. Probe accuracy trends reinforce this pattern, indicating that despite local geometric failures, large portions of each model preserve a shared, dataset-invariant emotional subspace.